

Harsath Kaliappa Gounder Thangavel

☎ +1 647-619-3864 ✉ harsath@fastmail.com 🐙 github.com/harsath 🌐 harsath.net 🔗 linkedin.com/in/harsath

Education

York University

Bachelor of Science in Computer Science (Honours)

January 2020 – December 2024

Toronto, Canada

- Data Structures
- Data Mining
- Computer Networking
- Computer Architecture
- Algorithm Design
- Database Systems
- Compiler Design
- Operating Systems

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Implemented REST APIs in Flask and integrated with Discord API to serve customers on Privex's Discord server.
- Utilized vector databases and MySQL for training data management, allowing dynamic updates from Discord by Privex staff.

CogniPet

Software Engineer

April 2019 – January 2020

Zürich, Switzerland

- Trained and deployed a deep learning model using the Xception neural network to recognize 120 breeds of dogs with 93% accuracy.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).

Projects

Jix Interpreter | *Interpreter written in C*

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented features such as fault-tolerant parsing and error handling, dynamic types (int, bool, string, and array), functions, nested blocks, recursion, built-in functions, if-else, return, and break statements, while and for loops.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.
- Tested the interpreter using automated tests written in the Jix source code and integrated with CI/CD pipelines.

Cap Containers | *C header-only containers library*

- Developed a highly-generic, self-contained, header-only containers library in C.
- Supported sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority), stack, lists, and an arena memory allocator.
- Included thread-safe variants with cross-platform support (Windows, Linux, and macOS).
- Tested using automated tests and CI/CD pipelines, and documented with Doxygen.

Blueth | *C++ Linux networking framework*

- Developed a networking framework for Linux providing common client-server abstractions for high-performance, concurrent, and event-driven networking services.
- Implemented event-loop abstraction, LRU cache, HTTP parsers and proxies, thread-pool executors, and codecs.

HashRing | *Go module*

- Implemented a consistent hashing data structure in Go using UUIDs and BLOBs.

SOCKS5 Handler | *SOCKS5 proxy client*

- Developed a C++ library implementing RFC 1928 and RFC 1929 for a SOCKS5 client with username/password authentication and remote DNS resolution; achieved 30 stars on GitHub.

Technical Skills

Languages: Python, C++, C, Java, Go, JavaScript, SQL, Perl, Bash, PHP, HTML/CSS, L^AT_EX
Technologies: Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, Neo4j
Developer Tools: Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services